MANNESMANN
LINE PIPE
Ein Unternehmen der Salzgitter Gruppe

**Inspection certificate 3.1**
**EN 10204:2004**
(A02)

No. (A03) **10001475**
Page 1/6
Date 02.01.2025

| | | | |
|---|---|---|---|
| No. (A07) Purchaser (A06) | PHP03478 15.10.2024<br>Salzgitter Mannesmann International (USA) Inc.<br>1770 St. James Place, Suite 600<br>77056-3499 HOUSTON<br>USA | No. (A07) Customer (A06) | Industrial Piping Specialists<br>Box 581270<br>74158 TULSA<br>USA |

Product (B01): steel line pipe, HFW-HFI-longitudinally welded

Works order No. (A08): 0603010573
Dispatch note No.: 0080030793
10.12.2024
Inspection (A05): Manufacturer acc.

Steel grade and terms of delivery (B02-B03):
API SPEC 5L 46TH PSL 2 X65M
Hardness acc. to NACE MR0175 / ISO15156
Hardness acc. to NACE MR0103
Hydrostatically tested 100% SMYS
Impact Requirements according to API 5L 46th: Base material at -29°C (M20°F), HAZ at -29°C (M20°F) and Weld seam at -20°C (M4°F) / shear area determination base material for information only.
marking acc. to API 5L 46th

Heat treatment: Weld zone inductive annealed
(Temperature minimum 1600°F / 871°C)

## Material data (B01-B99)

| Item | Heat No. (B07) | Quantity (B08) | Cond. of delivery (B04) | D x T x Length (B09-B11) in x in x ft | Length (B09-B11) ft | Weight (B12) lbs |
|---|---|---|---|---|---|---|
| 040 | 730213 | 2 | U | 16,000 x 0,500 x 44,967 | 89,783 | 7.438 |
| 040 | 761641 | 58 | U | 16,000 x 0,500 x 44,967 | 2.598,247 | 215.254 |
| Σ | | 60 | U: as rolled | | 2.688,030 | 222.692 |

The mentioned length and weight refers to the product in delivery condition

## Product description (B06)

Inside conditions (B06)
Surface: black

Outside conditions (B06)
Surface: black temporary anti-corrosive protection laquer approx. 35 my thickness

ends (B06)
End 1: bevelled 30 +5/-0°, root face 1,6 +/-0,8 mm
End 2: bevelled 30 +5/-0°, root face 1,6 +/-0,8 mm
End protection: without plastic caps

Marking of the product (B06)
Longitudinal marking at one end:
Manufacturers sign SIEGEN API SPEC 5L 0035 API monogram
MM-YY 16.000 0.500 X65M PSL 2 HFW
A HEAT NO. Heat-no. Single pipe length (ft) Pipe-no. TESTED 3840 PSI
MADE IN GERMANY IPS P.O. No. PHP03478

Internal marking at one end:
Barcode Pipe-no.
Product No. Manufacturers sign
Single pipe length (ft) HEAT No. Heat-no.
TESTED 3840 PSI IPS

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## Ladle analysis (C70–C99)
Manufacturer standard

| Heat No. (B07) | C % ≤0,10 | Si % ≤0,45 | Mn % ≤1,70 | P % ≤0,025 | S % ≤0,015 | Cu % ≤0,50 | Cr % ≤0,50 | Ni % ≤0,50 | Mo % ≤0,50 | V % |
|---|---|---|---|---|---|---|---|---|---|---|
| 730213 | 0,07 | 0,32 | 1,22 | 0,011 | 0,002 | 0,02 | 0,03 | 0,02 | 0,003 | 0,002 |
| 761641 | 0,07 | 0,32 | 1,23 | 0,009 | 0,002 | 0,02 | 0,03 | 0,02 | 0,002 | 0,002 |

| Heat No. (B07) | Ti % | Nb % | B % ≤0,0010 | EV1 1) % ≤0,15 |
|---|---|---|---|---|
| 730213 | 0,02 | 0,04 | 0,0002 | 0,06 |
| 761641 | 0,02 | 0,04 | 0,0001 | 0,06 |

1) EV1 : Nb+V+Ti

Steelmaking process: Basic oxygen process
(C70)

Steel Manufacturer incl. country of melt and pour (A01):
Hüttenwerke Krupp Mannesmann GmbH, Postfach 25 11 24, 47251 DUISBURG, GERMANY

Coil Manufacturer (A01):
ThyssenKrupp Steel Europe AG, Postfach 4 71 61, 47166 DUISBURG, GERMANY

## Product analysis (C70–C99)
ISO/TR 9769

Test type: Lot testing

| Specimen No. (C00) | Heat No. (B07) | C % ≤0,10 | Si % ≤0,45 | Mn % ≤1,70 | P % ≤0,025 | S % ≤0,015 | Cu % ≤0,50 | Cr % ≤0,50 | Ni % ≤0,50 |
|---|---|---|---|---|---|---|---|---|---|
| 1002975345 *) | 730213 | 0,05 | 0,33 | 1,23 | 0,010 | 0,001 | 0,02 | 0,03 | 0,02 |
| 1002975346 *) | 730213 | 0,05 | 0,33 | 1,22 | 0,009 | 0,001 | 0,02 | 0,03 | 0,02 |
| 1002975278 | 761641 | 0,05 | 0,33 | 1,24 | 0,008 | 0,002 | 0,02 | 0,03 | 0,02 |
| 1002975279 | 761641 | 0,05 | 0,33 | 1,25 | 0,009 | 0,002 | 0,02 | 0,03 | 0,02 |

| Specimen No. (C00) | Heat No. (B07) | Mo % ≤0,50 | V % | Ti % | Nb % | B % ≤0,0010 | EV1 1) % ≤0,25 | EV2 2) % ≤0,15 |
|---|---|---|---|---|---|---|---|---|
| 1002975345 *) | 730213 | 0,01 | 0,002 | 0,02 | 0,04 | 0,0001 | 0,13 | 0,07 |
| 1002975346 *) | 730213 | 0,01 | 0,002 | 0,02 | 0,04 | 0,0001 | 0,12 | 0,07 |
| 1002975278 | 761641 | 0,01 | 0,001 | 0,02 | 0,05 | 0,0001 | 0,13 | 0,07 |
| 1002975279 | 761641 | 0,01 | 0,001 | 0,02 | 0,05 | 0,0001 | 0,13 | 0,07 |

1) EV1 : PCM = C+Si/30+(Mn+Cu+Cr)/20+Mo/15+Ni/60+V/10+5*B
2) EV2 : Nb+V+Ti
*) The sample pipe is not part of the delivery

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## Tensile test (C10–C29)
ISO 6892–1

Test type: Lot testing
Condition specimen: Specimen as delivery condition/Pipe as rolled
Specimen shape: Flat tensile test

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Height of specimen in | Width of specimen in | Yield strength (C11) Rt0,5 psi 65300–87000 | Tensile strength (C12) Rm psi 77600–110200 | Rt0,5/Rm ≤0,93 |
|---|---|---|---|---|---|---|---|---|
| 1002975278 | 761641 | BM | T180 | 0,51 | 1,57 | 72519 | 89488 | 0,81 |
| 1002975345 *) | 730213 | BM | T180 | 0,51 | 1,57 | 76145 | 93114 | 0,82 |

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Elongation (C13) 2in % ≥23 |
|---|---|---|---|---|
| 1002975278 | 761641 | BM | T180 | 37 |
| 1002975345 *) | 730213 | BM | T180 | 39 |

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Height of specimen in | Width of specimen in | Tensile strength (C12) Rm psi 77600–110200 |
|---|---|---|---|---|---|---|
| 1002975278 | 761641 | WL | T000 | 0,51 | 1,57 | 81511 |
| 1002975345 *) | 730213 | WL | T000 | 0,51 | 1,57 | 81946 |

1) BM: Base material
WL: Weld seam
2) T000: Transversal 0°
T180: Transversal 180°
*) The sample pipe is not part of the delivery

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## Charpy impact test (C40–C49)
ASTM A 370

Test type: Lot testing
Condition specimen: Specimen as delivery condition/Pipe as rolled
Type of notch: V–notch
Specimen size: 1/1 specimen

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Temp. (C03) °F | Impact energy (C42–C43) 1 ftlbf ≥15 | 2 | 3 | MW 3) ftlbf ≥20 |
|---|---|---|---|---|---|---|---|---|
| 1002975278 | 761641 | BM | T090 | −020 | 151 | 155 | 150 | 152 |
| 1002975345 *) | 730213 | BM | T090 | −020 | 173 | 169 | 170 | 171 |

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Temp. (C03) °F | Shear fracture (C42–C43) 1 % | 2 | 3 | MW 3) % |
|---|---|---|---|---|---|---|---|---|
| 1002975278 | 761641 | BM | T090 | −020 | 90 | 90 | 90 | 90 |
| 1002975345 *) | 730213 | BM | T090 | −020 | 90 | 90 | 90 | 90 |

| Specimen No. (C00) | Heat No. (B07) | Location (C01) 1) | Orient. (C02) 2) | Temp. (C03) °F | Impact energy (C42–C43) 1 ftlbf ≥15 | 2 | 3 | MW 3) ftlbf ≥20 |
|---|---|---|---|---|---|---|---|---|
| 1002975278 | 761641 | HAZ | T000 | −020 | 181 | 185 | 185 | 184 |
| 1002975278 | 761641 | WL | T000 | −004 | 177 | 165 | 162 | 168 |
| 1002975345 *) | 730213 | HAZ | T000 | −020 | 181 | 184 | 181 | 182 |
| 1002975345 *) | 730213 | WL | T000 | −004 | 218 | 168 | 219 | 202 |

1) BM: Base material
HAZ: Heat affected zone
WL: Weld seam
*) The sample pipe is not part of the delivery

2) T000: Transversal 0°
T090: Transversal 90°
3) MW: Average

## Hardness test (C30–C39)
ISO 6507–1

Test type: Lot testing

| Specimen No. (C00) | Heat No. (B07) | Location 1) | Härtetyp 2) | Hardness max. HV10 ≤250 | | | | |
|---|---|---|---|---|---|---|---|---|
| 1002975278 | 761641 | | HV | 218 | | | | |
| 1002975345 *) | 730213 | | HV | 226 | | | | |

2) HV: hardness Vickers
*) The sample pipe is not part of the delivery

## Destructive tests (D51–D99)

| | |
|---|---|
| Flattening test according to ISO 8492: | Satisfactory |
| Metallographic determination of micro and macro structure: | Satisfactory |

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## Non–destructive tests (D02–D50)

| | |
|---|---|
| Visual and dimensional inspection: | Satisfactory |
| Ultrasonic testing of weld seam for the detection of longitudinal imperfections according to ISO 10893–11, acceptance level U3: | Satisfactory |
| Hydrostatic test 3840 PSI, holding time 5 s: | Satisfactory |
| Check for residual magnetism at the pipe ends max. 30 Gauss: | Satisfactory |

## Pipe list (B01–B99)

| Item | Pipe No. (B07) | Product No. (B07) | Heat No. (B07) | Inspection No. (B07) | Cond. of delivery (B04) | Length ft (B09–B11) | Weight lbs (B12) |
|---|---|---|---|---|---|---|---|
| 040 | 1002975274 | 25450101 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975275 | 25450102 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975276 | 25450103 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975277 | 25450104 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975278 | 25450105 | 761641 | 0 | U | 44,869 | 3.717 |
| 040 | 1002975279 | 25450106 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975280 | 25450107 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975281 | 25450108 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975282 | 25450109 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975283 | 25450110 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975284 | 25450111 | 761641 | 0 | U | 44,902 | 3.721 |
| 040 | 1002975285 | 25450112 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975286 | 25450113 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975288 | 25450201 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975289 | 25450202 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975290 | 25450203 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975291 | 25450204 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975292 | 25450205 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975293 | 25450206 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975294 | 25450207 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975295 | 25450208 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975296 | 25450209 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975297 | 25450210 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975298 | 25450211 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975299 | 25450212 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975300 | 25450213 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975302 | 25450301 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975303 | 25450302 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975304 | 25450303 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975305 | 25450304 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975306 | 25450305 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975307 | 25450306 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975308 | 25450307 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975309 | 25450308 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975310 | 25450309 | 761641 | 0 | U | 44,902 | 3.721 |
| 040 | 1002975312 | 25450401 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975313 | 25450402 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975314 | 25450403 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975315 | 25450404 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975316 | 25450405 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975317 | 25450406 | 761641 | 0 | U | 44,882 | 3.719 |
| 040 | 1002975319 | 25450408 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975320 | 25450409 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975321 | 25450410 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975322 | 25450411 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975323 | 25450412 | 761641 | 0 | U | 44,902 | 3.721 |

U : as rolled

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Date 02.01.2025

## Pipe list (B01–B99)

| Item | Pipe No. (B07) | Product No. (B07) | Heat No. (B07) | Inspection No. (B07) | Cond. of delivery (B04) | Length ft (B09–B11) | Weight lbs (B12) |
|---|---|---|---|---|---|---|---|
| 040 | 1002975324 | 25450413 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975326 | 25450501 | 761641 | 0 | U | 44,892 | 3.719 |
| 040 | 1002975327 | 25450502 | 761641 | 0 | U | 44,895 | 3.719 |
| 040 | 1002975328 | 25450503 | 761641 | 0 | U | 44,885 | 3.719 |
| 040 | 1002975329 | 25450504 | 761641 | 0 | U | 44,902 | 3.721 |
| 040 | 1002975330 | 25450505 | 761641 | 0 | U | 44,882 | 3.719 |
| 040 | 1002975333 | 25450508 | 761641 | 0 | U | 39,406 | 3.265 |
| 040 | 1002975334 | 25450509 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975335 | 25450510 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975336 | 25450511 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975337 | 25450512 | 761641 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975338 | 25450513 | 761641 | 0 | U | 44,898 | 3.719 |
| 040 | 1002975340 | 25450601 | 730213 | 0 | U | 44,888 | 3.719 |
| 040 | 1002975351 | 25450612 | 730213 | 0 | U | 44,895 | 3.719 |
| Quantity (B08) 60 | | U: as rolled | | | | 2.688,030 | 222.692 |

Statement of compliance:
The products have been tested in accordance with the purchase specification and are found to be satisfactory. Alteration of this testimonial and certification or use for other products is not allowed and are regarded as falsification of documents and will be subjected to criminal jurisdiction.

We hereby certify that the delivered material complies with the terms of the order.
(Z01)

Trademark
(A04)

**Mannesmann Line Pipe GmbH**
In der Steinwiese 31
57074 SIEGEN
GERMANY
(A01)

A Inspection Stamp (Z03)

Inspection Representative (Z02)
Sascha Arndt

# Production Bend Report

Tulsa Tube Bending

| | | | |
|---|---|---|---|
| **Customer:** | DNOW | **Report No.:** | 109710-PBR-1 |
| **Customer PO No.:** | 663013318 | **Code of Fabrication:** | |
| **TTB Job No.:** | 26H109710 | B16.49-2023/42414SP | |

**Mother Pipe & Bend Information**

OD (in): 16.000 Grade: API-5L-X65 PSL 2 Radius (in): 80 Bending Machine ID: M2 Heat No.: 761641

**Essential Variables**

Wall Thickness (inch): 0.500 Radius-to-Diameter Ratio: 5.0 Coil Design: TTB High Spec Coolant Type: Water

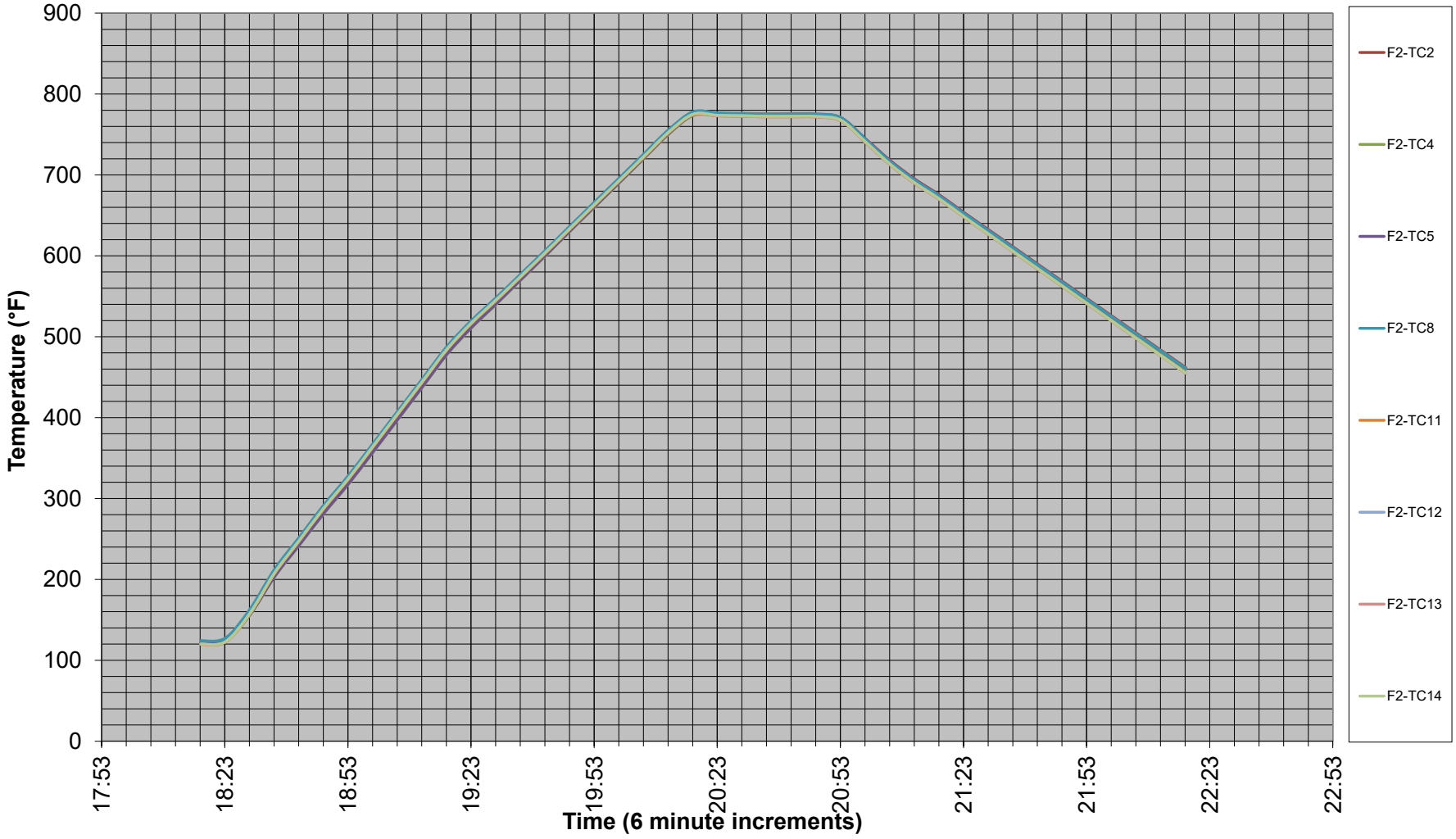
Weld Seam Location: 0 degrees Heat Treatment Holding Temp (°F): 775 Heat Treatment Soak Time (min): 30.0

**Note 1:** Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

| Bend ID No. | Operator Initials | Date | Bending Speed (ipm) | Extrados Bend Temp (°F): | Intrados Bend Temp (°F): | Coolant Temp (°F) | Coolant Flow Rate (GPM) | Induction Frequency (Hz) |
|---|---|---|---|---|---|---|---|---|
| Qualification Bend Value: | | | See Note 1. | | | 80.4 - 82.8 | 11.1 | 803 |
| Limits of Variation: | | | +/- 0.1 ipm | +/- 50°F | +/- 50°F | +/- 25°F | +/- 10% | +/- 20% |
| 109710-1-1 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.02 - 86.54 | 11.4 | 804 |
| 109710-1-2 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.38 - 92.66 | 11.4 | 802 |
| 109710-1-3 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.2 - 87.8 | 11.4 | 803 |
| 109710-1-4 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.38 - 85.28 | 11.5 | 803 |
| 109710-2-1 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 86.9 - 89.78 | 11.4 | 805 |
| 109710-2-2 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 79.34 - 88.88 | 11.2 | 804 |
| 109710-2-3 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 81.5 - 85.28 | 11.7 | 805 |
| 109710-2-4 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 82.76 - 85.28 | 11.4 | 805 |
| 109710-3-1 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.74 - 87.62 | 11.4 | 805 |
| 109710-3-2 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.56 - 93.38 | 11.2 | 806 |
| 109710-3-3 | JC | 7/30/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.56 - 93.38 | 11.2 | 806 |
| 109710-3-4 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 81.14 - 83.48 | 11.3 | 806 |
| 109710-3-5 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 82.4 - 84.56 | 11.6 | 806 |
| 109710-3-6 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 83.12 - 86.18 | 11.8 | 806 |
| 109710-3-7 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 83.66 - 85.82 | 11.1 | 806 |
| 109710-3-8 | JC | 7/31/2026 | +/- 0.1 ipm | +/- 50°F | +/- 50°F | 84.56 - 87.26 | 11.6 | 805 |

| | | | |
|---|---|---|---|
| **Customer** | **Tulsa Tube Bending Company** | **Zentech Job 1-30-26-0021096** | **8/5/2026** |
| **Job** | **26H109710-1** | | |
| **PO** | **26H109710-1** | | |
| **Part No.** | **Multi** | | |

## Zentech Thermal Processing
## Furnace No. F-2




| | | | | |
|---|---|---|---|---|
| **Calibration Date:** | **7/1/2026** | | | **Zentech Operator** |
| **Calibration Due:** | **10/1/2026** | **Thermocouple Type :** | **K** | **Signature :** |

# GLOBE X-RAY SERVICES

**8441 South Union Tulsa, OK. 74132**

Certification: 197451-1

## Report Of Ultrasonic Examination

Customer: Tulsa Tube Bending Job # 26H109710-1

Address:

Description:

The items listed below have been ultrasonically examined and meet the requirements in accordance with the attached calibration.

Unless otherwise noted by an * following the joint number, there were no recordable indications observed.

| Description | Joint No. | Accept | Reject |
|---|---|---|---|
| Long Seam | 109710-1-1,109710-1-2,109710-1-3,109710-1-4 | √ | |
| Long Seam | 109710-2-1,109710-2-2,109710-2-3,109710-2-4 | √ | |
| Long Seam | 10710-3-1,10710-3-2,10710-3-3,10710-3-4 | √ | |
| Long Seam | 10710-3-5,10710-3-6,10710-3-7,10710-3-8 | √ | |
| | | | |
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| | | | |
| | | | |

Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF ANY KIND DUE TO ANY KIND OF INTERPRETATION

# Dimensional-Visual Inspection Report

Tulsa Tube Bending

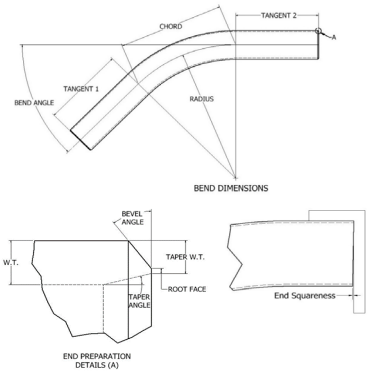
| | | | | | |
|---|---|---|---|---|---|
| **Customer:** | DNOW | **Pipe Size (in):** | 16 x 0.500 | **Report No.:** | 109710-DV-2-4 |
| **Customer PO No.:** | 663013318 | **Grade:** | API-5L-X65 PSL 2 | **Bend ID No.:** | 109710-2-4 |
| **TTB Job No.:** | 26H109710 | **Heat No.:** | 761641 | **Code:** | B16.49-2023/42414SP |

**Physical Dimensions**

| Dimension | Target | Tolerance | Actual |
|---|---|---|---|
| Tangent 1 (in) | 24.0 | +/- 0.19 in | 24.00 |
| Tangent 2 (in) | 24.0 | +/- 0.19 in | 23.88 |
| Chord (in) | 41.4 | n/a | 41.0 |
| Angle (deg) | 30.0 | +/- 0.5 ° | 29.8 |
| Radius, R (in) | 80 | +/- 1% | 79.7 |
| **Performed By:** | DW | **Date:** | 8/17/2026 |

**End Prep Dimensions**

| Dimension | Target | Tolerance | Actual | |
|---|---|---|---|---|
| Bevel Angle (deg) | 32.5 | +/- 2.5° | 32.50 | END 1 |
| Root Face (in) | 0.06 | +/- 0.03 in | 0.06 | |
| Taper Angle (deg) | n/a | n/a | n/a | |
| Min ID | 15 | +/- 0.1 in | 14.94 | |
| Max ID | 15 | +/- 0.1 in | 15.00 | |
| Max End Sq. (in) | 0 | +/- 0.09 in | 0.06 | |
| Bevel Angle (deg) | 32.5 | +/- 2.5° | 32.50 | END 2 |
| Root Face (in) | 0.06 | +/- 0.03 in | 0.06 | |
| Taper Angle (deg) | n/a | n/a | n/a | |
| Min ID | 15 | +/- 0.1 in | 15.00 | |
| Max ID | 15 | +/- 0.1 in | 15.00 | |
| Max End Sq. (in) | 0 | +/- 0.09 in | 0.07 | |
| **Performed By:** | DW | **Date:** | 8/17/2026 | |




**Ovality** **[(O.D. max – O.D. min)/O.D. nominal] x 100**

| Quadrant | Limit | Min OD | Max OD | % Ovality |
|---|---|---|---|---|
| End 1 | 1.0% | 15.958 | 15.976 | 0.1% |
| Start of Bend | 3.0% | 15.924 | 15.936 | 0.1% |
| 1/4 of Bend Arc | 3.0% | 15.911 | 15.938 | 0.2% |
| 1/2 of Bend Arc | 3.0% | 15.910 | 15.942 | 0.2% |
| 3/4 of Bend Arc | 3.0% | 15.926 | 15.957 | 0.2% |
| Stop of Bend | 3.0% | 15.905 | 16.010 | 0.7% |
| End 2 | 1.0% | 15.961 | 16.089 | 0.8% |
| **Performed By:** | DW | | **Date:** | 8/17/2026 |

**Wall Thinning of Extrados**

| Quadrant | Limit | Nominal Wall Thickness | Extrados Wall Thickness | % Wall Thinning |
|---|---|---|---|---|
| Start of Bend | 10.0% | 0.500 | 0.480 | 4.0% |
| 1/4 of Bend Arc | 10.0% | 0.500 | 0.476 | 4.8% |
| 1/2 of Bend Arc | 10.0% | 0.500 | 0.472 | 5.6% |
| 3/4 of Bend Arc | 10.0% | 0.500 | 0.472 | 5.6% |
| Stop of Bend | 10.0% | 0.500 | 0.489 | 2.2% |
| **Performed By:** | DW | | **Date:** | 8/17/2026 |

**ID Gauge Drift/Pig Inspection**

| | | | |
|---|---|---|---|
| Minimum Gauge Diameter = | 97.0% | of Nominal ID, or | 14.550 in |
| Gauge Length Required = | 15 in | | |

Gauge passes through the entire bend without power equipment assistance

| | | | |
|---|---|---|---|
| **Performed By:** | DW | **Date:** | 8/17/2026 |

**Visual Inspection**

All accessible surfaces found to be free of visible injurious defects.
Bend is stenciled with correct information.

| | | | |
|---|---|---|---|
| **Performed By:** | DW | **Date:** | 8/17/2026 |

**READINGS:** ✓ MEET ACCEPTANCE CRITERIA

☐ DO NOT MEET ACCEPTANCE CRITERIA

# GLOBE X-RAY SERVICES

**8441 South Union Tulsa, OK. 74132**

Certification: 197451-1

## UT Examination Record - Calibration Data Report

Customer: Tulsa Tube Bending Date: August 13, 2026

Job # 26H109710-1 Reference Code/Specification : ASME sec. V, API5L

Procedure # GXS-II-F Rev No. 1-26 Acceptance Standard: ASME sec. VIII, API5L

Instrument: Olympus Epoch 650 S/N: 231803608 Couplant : x Sonix ___ Other: ___

Cable Type: CO-Axial RG-174 x 6' Long

Method:

[X] Contact [X] Manual Scan [X] Pulse Echo [X] Single Element [ ] Dual Element [ ] Array

## General Calibration ( Systems Cal.) Prior to Examination

| Search Unit | | | | | | | Instrument Settings | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Search Unit | Beam Angle | Size | Frequency MHz | S/N | Wave Form Shear | Wave Form Longitude. | Damping | Range | Delay | Reject | Frequency MHz | Init.Cal Check | Cal Check | Final Cal Check |
| #1 | 0° | 1" | 2.25 MHz | U12708 | | X | 50 | 2.5 | 1.07 | 0 | 2.25 MHz | 1230 | | 1300 |
| #2 | 70° | 1/2" | 2.25 MHz | SE0679 | X | | 400 | 5 | 9.27 | 0 | 2.25 MHz | 1300 | | 1400 |
| #3 | 45° | 1/2" | 2.25 MHz | SE0646 | X | | 400 | 5 | 10.88 | 0 | 2.25 MHz | 1400 | | 1500 |
| #4 | | | | | | | | | | | | | | |

## Calibration Block Identity

Unless noted, Beam Angle, Resolution, Beam Exit Point, and Sound Path Calibration were Determined on an IIW Block Type II

Serial # 94-5907 Material Type P-1

| Type | Serial Number | Material Type | Temp | Reflector | Ref. Gain | Ref. Ind. Amplitude | DAC When Used Electric | DAC When Used Manual | DAC Range When Used | Search Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| ASME Curve | P1-14/80-1 | P-1 | Amb. | 3/32" SDH | 57.5 Db | 80 %FSH | X | | 2.92 S.P. | #2 |
| ASME Curve | P1-14/80-1 | P-1 | Amb. | 3/32" SDH | 56.6 Db | 80 %FSH | | | 1.41 S.P. | #3 |
| | | | | | | | | | | |
| | | | | | | | | | | |

Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF ANY KIND DUE TO ANY KIND OF INTERPRETATION

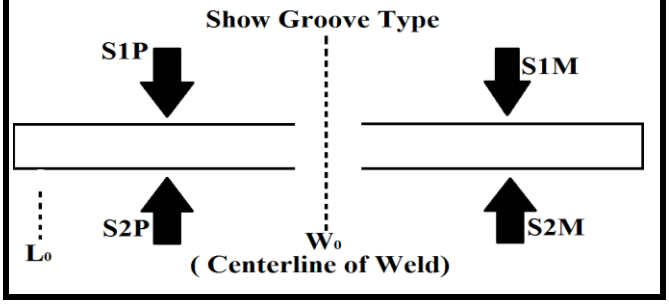
# GLOBE X-RAY SERVICES

**8441 South Union Tulsa, OK. 74132**

## Longitudinal Welds Legend

| Exam | Scan Number | Type of Scan | Surface Number |
|---|---|---|---|
| X | 1 | Straight Beam Scan – lamination exam prior to weld exam | S1 |
| X | 2 | Angle Beam Scan Perpendicular to the Weld from Surface | S-1P |
| X | 3 | Angle Beam Scan Perpendicular to the Weld from Surface | S-1M |
| | 4 | Angle Beam Scan Perpendicular to the Weld from Surface | S-2P |
| | 5 | Angle Beam Scan Perpendicular to the Weld from Surface | S-2M |
| X | 6 | Angle Beam Scan Parallel to the Weld from the Surface | S-1P |
| X | 7 | Angle Beam Scan Parallel to the Weld from the Surface | S-1M |
| | 8 | Angle Beam Scan Parallel to the Weld from the Surface | S-2P |
| | 9 | Angle Beam Scan Parallel to the Weld from the Surface | S-2M |
| | 10 | Straight Beam Scan- Weld Exam, Surface | S-1P & S-1M, Weld Ground Flush |
| | 11 | Straight Beam Scan- Weld Exam, Surface | S-2P & s-2M, Weld Ground Flush |

| Transducer Coordinates | |
|---|---|
| $L_0$ | 0° or 0 inch of Nozzle Layout (Starting Point for Weld Layout) |
| $L_1$ | Distance from $L_0$, in Clockwise Direction to Centerline of Transducer at 25% signal amplitude, near end of Indication. |
| $L_{Max}$ | Distance from $L_0$, in Clockwise Direction to Centerline of Transducer at Max Signal Amplitude of Indication. |
| $L_2$ | Distance from $L_0$, in Clockwise Direction to centerline of transducer at 25% signal amplitude, at far end of Indication. |
| $W_0$ | Centerline of Weld |
| $W_{Max}$ | Distance from $W_0$, to Exit Point of Transducer at Max Signal Amplitude of Indication. |
| SP | Sound Path Distance to Indication |




Scanning Surface Identification:

**S1:** Vessel Outside Surface

S2: Vessel Inside Surface

**P:** Plus side of reference point $W_0$ ; Surface on the clockwise direction of the 0 Deg point as viewed from vessel top for vertical, vessel as viewed from the inlet/reference end for horizontal vessel.

**M:** Minus side of reference point $W_0$ ; Surface on the counter-clockwise direction of the 0 point as viewed from vessel top for vert-vessel as viewed from the inlet/reference end for horizontal vessel.

## Starting Point for Weld Layout

Longitudinal Weld:
- [ ] Vertical Vessel: Centerline of circumferential weld at top of longitudinal weld
- [X] Horizontal Vessel: Centerline of circumferential weld on inlet/reference end

Layout Increment Spacing: _____ Number of Layout Points: _____

Weld Accessibility:
- [X] 100% Accessible
- [ ] Restricted Access - Weld Examined with Angle Beam from one side of weld axis only.
- [ ] Inaccessible area(s)- Area of weld which cannot be examined from at least one side with angle beam.
- [ ] Inaccessible from increment number _____ to _____ / _____ to _____
  _____ to _____ / _____ to _____

Examination By: Colter Moyer Colter Moyer Qualification Level : II

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# GLOBE X-RAY SERVICES

**8441 South Union Tulsa, OK. 74132**

Certification: 197451-1

## UT Examination Record

Customer: Tulsa Tube Bending Job# 26H109710-1

Weld Identification: See cover page

Material Type: C.S. Material Thickness: .500"

Material Temperature: Ambient Material Diameter: 16"

## Test Conditions

Surface Condition of Weld: ____ Ground Flush ____ Ground Smooth X As Welded

Surface Condition of the Scan Surface: X Buffed Smooth ____ Other: ____

## Results of Ultrasonic Examination

Accept

No recordable indications noted.

See Attached Legend and Calibration record for the following data.

**Examination Scan Numbers this Examination: 1,2,3,6,7**

| Indication # | Scan No. | Search Unit No. | Beam Angle | % DAC | Search Unit Location | | Length Dimensions | | Reflector Depth | Reflector Length | Sound Path | Eval. | Results |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | W Max | L Max | L1 | L2 | | | | | |
| | | | | | | | | | | | | | |
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Examination By: Colter Moyer Qualification Level : II

Printed: Colter Moyer Date : August 13, 2026

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**Globe X-Ray Services, Inc.**
**8441 S. Union Ave.**
**Tulsa, OK 74132**
**918-446-1696**



**Certification # 195947**

# Magnetic Particle Examination Record

Customer: Tulsa Tube Bending Date: 8/13/2026

Job/P.O. No.: 109710 Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C Rev. No. 1/26 Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300 Serial No. 1861 Technique: X Yoke _____ Prods

Current Type: X A/C _____ D/C Magnetic Particle Type: _____ Wet X Dry _____ Fluorescent X Visible

Amperage (Prod Technique Only) _____ Amps/inch of Prod Spacing.

Part/Item Material Type: Carbon Material Thickness: Varies

Lighting Equipment: _____ Fluorescent Light Model # _____ Serial # _____
X White Light using Parker Research Y300 or Y400 Probe light
_____ Natural White Light
_____ Other: _____
Light Intensity on surface to be examined: 100 Min foot candles N/A µW/cm²

Comments: Test The Entire Extrados (Back Half) of the Bend. From N.A to N,A, Including Weld Seam
Verify Residual Magnetism at Each End is <25 Gauss

| Examination Results | | | | |
|---|---|---|---|---|
| Qty: | Part Identification | Accept | Reject | Indication Description |
| 16 | 16" O.D. X 0.5" Wall,<br>80" CLR, API-5L-X65 PSL 2 ERW Pipe<br>HT# 761641<br>Bend ID # 109710-1-1, 109710-1-2, 109710-1-3, 109710-1-4, 109710-2-1, 109710-2-2, 109710-2-3, 109710-2-4, 109710-3-1, 109710-3-2, 109710-3-3, 109710-3-4, 109710-3-5, 109710-3-6, 109710-3-7, 109710-3-8 | X | | None |

Examine: Jaime Vera Qualification Level: II

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**Tulsa Tube Bending**
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595
*Serving others. Building people. Pursuing excellence.*

# Certificate of Conformance

| | | | |
|---|---|---|---|
| Customer: | DNOW | Certificate No.: | 20260818 |
| Customer PO No.: | 663013318 | Issue Date: | 8/18/2026 |
| TTB Job No.: | 26H109710 | | |
| Drawing No.: | n/a | Revision: | 0 |
| Material Size, Grade: | 16.00 inch OD x 0.500 inch w.t., API-5L-X65 PSL 2 ERW | Quantity of Bends: | 16 |
| Material Heat No(s): | 761641 | | |

| | | | | | |
|---|---|---|---|---|---|
| Bend ID Nos.: | 109710-1-1 | 109710-1-2 | 109710-1-3 | 109710-1-4 | 109710-2-1 |
| | 109710-2-2 | 109710-2-3 | 109710-2-4 | 109710-3-1 | 109710-3-2 |
| | 109710-3-3 | 109710-3-4 | 109710-3-5 | 109710-3-6 | 109710-3-7 |

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

## B16.49-2023/42414SP

*With the following notes and exceptions:

(1) TTB is using welded HFW pipe instead of the SMLS pipe required by spec

(2) TTB will do charpy testing at +14F as per ASME B16.49 and not to -20F as required by the spec.

(3) TTB will only do UT of the weld seam (if welded HFW pipe is approved) and not RT as required by the spec.

(4) External Coating NOTES & EXCEPTIONS to Consumers Energy Coating Spec 26907sp per TTB Quote No. 269641.

Approved By: Austin L Carter Date: 8/18/2026

Keith INDUSTRIES
20701 FM 521 Rosharon , TX 77583
+1-713-991-5670

# Coating Inspection Record

KF-010 FBE 5/27/22 REV2

| | | |
|---|---|---|
| Customer: TTB | Date: 9/3/2026 | Items: 4 PCS |
| Customer PO: 26H 109710-1 | Paint Spec: CONSUMER ENERGY 26907SP | |

## Surface Preparation

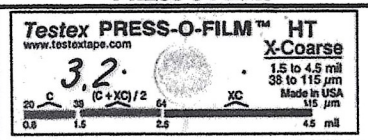
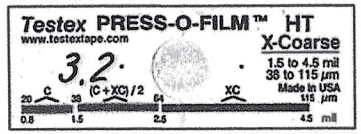
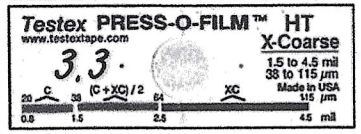
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Item Salt Test | N/A☐ | Result: <1 | SSPC SP-1 if needed | | Date | 9/3/2026 | 9/3/2026 |
| Abrasive Salt Test | N/A☐ | Result: <1 | Cleanliness Required | SSPC SP-10 | Time | 8:30AM | 10:30AM |
| Laboratory Testing | ☐Yes ☒No | | Cleanliness Achieved | ☒Yes☐NO | Air Temp | 83°F | 90°F |
| Type of Abrasive | Steel Grit | | Profile Required | 2.0-4.0 mils | Relative Humidity | 77% | 75% |
| Nozzle PSI | ☐ N/A 100 psi ☒ | | Profile Achieved | ☒Yes☐NO | Steel | 84˚F | 96˚F |
| Blotter Test | ☒PASS | | SSPC-VIS 1 Used | ☒Yes ☐No | Dew Point | 75˚F | 81˚F |
| Dust Free Surfaces | ☒YES ☐No | | Visual Inspection | ☒Yes ☐No | Stage of Operation | Pre-Blast | Pre-Coating |

## Coating Application

| | | | | |
|---|---|---|---|---|
| Coat 1st | Product Name: PipeClad 2000 | Batch: CH1876UE | Min Mils: 14 | Target Pre Coat Temp: 450 deg F |

## Final Testing

| | | | |
|---|---|---|---|
| HVST @125v/Mil; 2000v MIN PASS ☒ | Adhesion Knife Test: PASS ☒ | DFT Test: PASS ☒ | All Requirements Met: PASS ☒ |

| KI# | Description | Mil Average | Press O Films | KI# | Description | Mil Average |
|---|---|---|---|---|---|---|
| 2-1 | 16" X .500 30 DEG BEND | 18 |  | | | |
| 2-2 | 16" X .500 30 DEG BEND | 23 |  | | | |
| 2-3 | 16" X .500 30 DEG BEND | 19 | | | | |
| 2-4 | 16" X .500 30 DEG BEND | 25 |  | | | |
| | | | | | | |



Verified by Keith: _____